Domain: Cyber Security

SECURE-ME

Where identity meets context

Team Phoenix

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PROBLEM STATEMENT

Digital banking systems are increasingly being targeted by deepfake-based **impersonation attacks** due to the rapid development of AI technologies.

Attackers are able to:

- Make use of voice and face clones created by AI.
- Change the Video KYC procedures
- Avoid using conventional biometric authentication
- During high-value transactions, pretend to be authentic users.

EXISTING SYSTEM

- OTP-based authentication for login and transactions
- Static **biometric checks** (face / fingerprint matching)
- Basic device-level validation
- Rule-based, binary authentication decisions
- Relies on independent, single-factor validations

LIMITATIONS

- OTP confirms device or SIM, not the person
- Biometrics can be replayed or spoofed
- No single system verifies human authenticity
- Decisions are often binary, **not risk-adaptive**
- Lacks multi-signal correlation for impersonation detection

PROPOSED SOLUTION

- A **risk-based authentication framework** to verify real human presence during high-risk banking actions.
- **Active-Liveness Detection** via System Brightness to verify the physical presence of the user.
- Combines **face verification, and video-spoofing** detection to counter replay and AI-generated impersonation.
- Applies **device fingerprinting** to recognize familiar environments and detect anomalies.
- Introduces a **private, non-displayed gesture** as a step-up authentication factor, with a central risk engine aggregating all signals.

FEATURES

ACTIVE LIVENESS DETECTION VIA SYSTEM BRIGHTNESS

Verifies the user is physically present by changing screen brightness and analyzing how light reflects on their face in real time.

DEVICE FINGERPRINTING

Device fingerprinting uses non-intrusive device and environment attributes to probabilistically recognize and assess the trust level of a user's device.

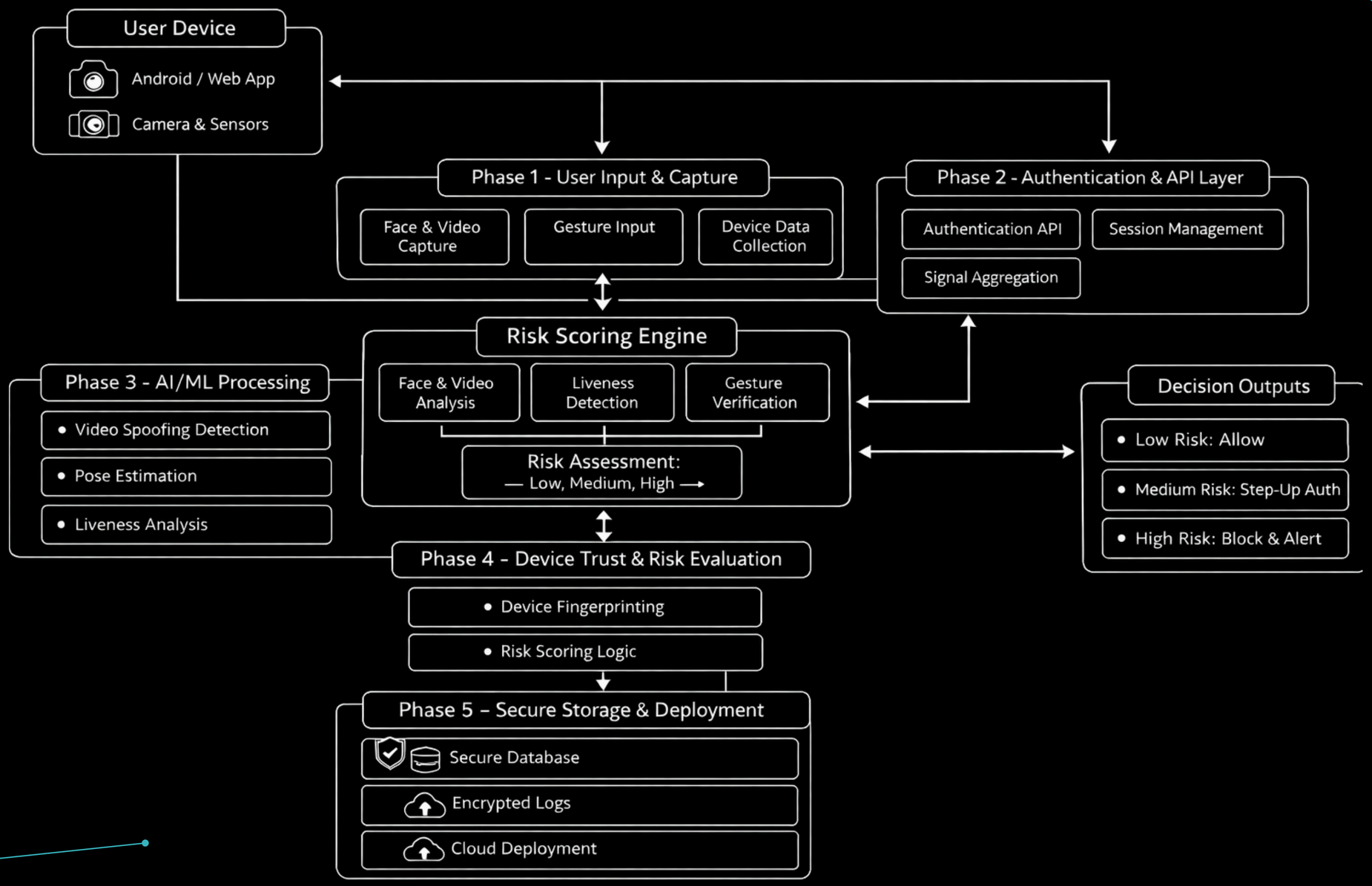
PRIVATE GESTURE AUTHENTICATION

A user-defined, non-displayed physical gesture used as a step-up authentication factor during high-risk scenarios

CENTRAL RISK SCORING ENGINE

Combines all security signals to assess risk and determine the final action.

SYSTEM ARCHITECTURE



TECH STACK

User Interface & Input

Android (Kotlin) / React, Camera APIs

Backend & APIs

Python (FastAPI / Flask), REST APIs

AI / ML Processing

Python, OpenCV, TensorFlow / PyTorch, MediaPipe

Device Trust & Risk Engine

Python, rule-based scoring models

Storage, Security & Deployment

PostgreSQL / MongoDB, AES, Docker, AWS/Azure

FEASIBILITY

- Uses existing smartphone cameras and sensors, **no extra hardware** needed.
- Built with **proven AI** and cloud technologies.
- Easily integrates with current banking systems via APIs.
- Supports **real-time authentication** with modern devices and infrastructure.

SCALABILITY

- **Cloud-based** architecture supports large numbers of users.
- **Modular design** allows easy feature expansion and upgrades.
- Can be deployed across multiple banks and digital platforms.
- Core authentication engine is **reusable** for other identity-critical systems.

IMPACTS

- Significantly reduces deepfake-driven identity fraud.
- Strengthens security of high-value digital banking transactions.
- Enhances reliability of video KYC and online authentication.
- Improves overall trust in digital financial platforms.

BENEFITS

- Prevents account takeovers and unauthorized access.
- Minimal friction for genuine users during authentication.
- Integrates easily with existing banking systems.
- Supports regulatory compliance and audit readiness.

FUTURE ENHANCEMENTS

Frame Interpolation Inconsistency

- Analyzes **motion continuity** across consecutive video frames during authentication.
- Compares observed motion patterns against natural human movement dynamics.
- Detects **unnatural transitions**, jitter, or over-smoothing caused by AI-generated or interpolated frames.
- Helps identify deepfake or synthetic videos that visually appear realistic but lack temporal consistency.

THANK
YOU